

A Law of Motion for Policy-Induced Changes in Unrealized Capital Gains

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1. Objective and Scope

This document specifies a principled accounting system for tracking the policy-induced change in unrealized capital gains in a repeated-cross-section microsimulation. The system supports revenue and behavioral analysis of proposals that alter (i) the capital gains tax rate and (ii) the treatment of unrealized gains at death.

The simulation contains cross-sectional records indexed by year and age. It does not follow individuals longitudinally. Nevertheless, the stock of unrealized capital gains evolves dynamically: gains held by a person aged a in year t shape what is held by that same person aged $a + 1$ in year $t + 1$. The accounting system tracks this dynamic at the age $\times$ year level by constructing a state variable that propagates across age cells and across years, treating each (age, year) cell as a synthetic cohort.

Scope

In scope:

- A law of motion for the policy-induced delta in unrealized gains, indexed by age, year, and asset class.
- A multiplicative realization-rate function with age, asset, regime, and tax-rate dependence.
- A regime taxonomy that supports step-up basis, carryover basis, and deemed realization at death, formulated through generic regime hooks so that additional regimes can be added without rederiving the recurrence.
- A ratio-based empirical anchoring that guarantees baseline reproduction.
- A calibration posture that distinguishes parameters drawn from data, calibrated to literature targets, and held as tunable assumptions.

Out of scope:

- The Haig-Simons income measure and accrual-based distributional analysis.
- Mark-to-market regimes that tax accruals annually.
- Charitable dispositions of appreciated assets, which bleed the unrealized stock without going through the realization channel.
- Forward-looking expectations of future tax rates; the model is myopic.

These are acknowledged in §9 as known gaps.

2. Notation and Primitives

2.1 Indices

- $a \in \{a_{\min}, \dots, a_{\max}\}$ — age of the tax unit; the topcoded cell $a_{\max}$ collects all ages at and above the topcode.
- t — year of the simulation.
- $k \in \mathcal{K} = \{\text{equities, pass_throughs, primary_home, other_home, re_fund}\}$ — asset class for which a basis exists and unrealized gains can be tracked.

2.2 Stocks and flows from data (baseline)

Each (a, t, k) cell carries the following baseline quantities, taken from the simulator’s baseline pass:

- $G_{a,t,k}^B$ — baseline unrealized taxable gain stock for the cell, defined as $\sum_i \max(0, V_{i,k} - B_{i,k})$ summed over tax units i in the cell.
- $r_{a,t,k}^B$ — baseline realization rate for the cell, defined as the ratio of baseline realized gains to $G_{a,t,k}^B$.
- $m_{a,t}$ — mortality rate. Probability that a tax unit at (a, t) ceases to exist as such by year $t + 1$ (for joint filers, the probability both spouses die; widow continuation is handled by the cross-section’s own re-imputation in subsequent years).

2.3 Allocation operators

- $A_{a \rightarrow h, t}$ — aging transition. Share of cell- (a, t) survivors reassigned to age h in year $t + 1$. Default: $A_{a \rightarrow a+1, t} = 1$ for $a < a_{\max}$; topcode loop $A_{a_{\max} \rightarrow a_{\max}, t} = 1$. Generalizes cleanly if a future variant uses fractional aging or finer age groups.
- $\omega_{a \rightarrow h, t}$ — heir-allocation matrix. Share of carryover gains from decedents at age a assigned to heirs at age h in year $t + 1$. Required to be row-stochastic in h :

$$\sum_h \omega_{a \rightarrow h, t} = 1 \quad \text{for every } a.$$

ω is supplied by the estate module and is regime-indifferent in its construction; its effect on the bathtub is gated by the regime hooks introduced in §3.3.

2.4 Policy primitives

- $\tau_{a,t}$ — capital gains marginal tax rate applicable to a cell.
- $c(\phi) \in [0, 1]$ — *post-death tax burden share*. The fraction of the embedded capital-gains tax that survives the holder’s death and is internalized in their pre-death realization decision. Canonical values: $c = 0$ under step-up (the gain is forgiven), $c = 1$ under deemed realization (the full tax is paid at death), $c \in (0, 1)$ under carryover (the heir eventually pays, scaled

by the holder's bequest motive). c enters the holder's lifetime lock-in calculation in §4. The downstream *routing* of decedent gains at the death event is handled separately by regime hooks ($\delta_{\text{vanish}}, \delta_{\text{route}}, \delta_{\text{realize}}$) defined in §3.3.

2.5 Reform-side stock and the policy-induced delta

Under a reform scenario, the unrealized gain stock evolves to $G_{a,t,k}^S$ and the realization rate to $r_{a,t,k}^S$. The state variable of interest is the **policy-induced delta**:

$$\Delta G_{a,t,k} = G_{a,t,k}^S - G_{a,t,k}^B.$$

By construction, $\Delta G_{a,t,k} = 0$ in the first year of any scenario; the delta accumulates over time as the reform diverges from baseline. Accruals on existing assets are assumed policy-invariant across scenarios in this document, so they difference out and do not appear in the recurrence for ΔG . (See §9.)

3. Law of Motion

3.1 Setup

The recurrence for $\Delta G_{h,t+1,k}$ is built by writing the reform-side recurrence for $G_{h,t+1,k}^S$ and the baseline-side recurrence for $G_{h,t+1,k}^B$ and differencing. Each year's stock at destination cell $(h, t+1, k)$ is composed of three sources:

1. **Survivors who aged into h .** Tax units at (a, t) who survived to year $t + 1$ and were reassigned to age h via A .
1. **Decedents whose gains transferred to heirs at h .** Tax units at (a, t) who died, with their gains routed to age h via ω — but only under regimes where carryover treatment applies.
1. **Accruals on the surviving stock.** Treated as policy-invariant in this model (see §9) and therefore not contributing to ΔG .

We treat the three components in order.

3.2 Survivor Flow

Consider the population at (a, t) . A fraction $1 - m_{a,t}$ survives to year $t + 1$. Of this surviving fraction, a share $r_{a,t,k}^S$ realizes their gains during year t , leaving a fraction $1 - r_{a,t,k}^S$ of the surviving stock to roll forward. The stock that arrives at destination cell h as survivors from age a is

$$G_{h,t+1,k}^{S,\text{surv}} = \sum_a A_{a \rightarrow h,t} (1 - m_{a,t}) (1 - r_{a,t,k}^S) G_{a,t,k}^S.$$

The corresponding baseline-side expression replaces r^S with r^B and G^S with G^B . Differencing and substituting $G_{a,t,k}^S = G_{a,t,k}^B + \Delta G_{a,t,k}$:

$$\Delta G_{h,t+1,k}^{\text{surv}} = \sum_a A_{a \rightarrow h,t} (1 - m_{a,t}) \left[(1 - r_{a,t,k}^S) G_{a,t,k}^S - (1 - r_{a,t,k}^B) G_{a,t,k}^B \right].$$

Expand the bracket using $G^S = G^B + \Delta G$:

$$\begin{aligned} (1 - r^S) G^S - (1 - r^B) G^B &= (1 - r^S)(G^B + \Delta G) - (1 - r^B) G^B \\ &= (1 - r^S) \Delta G + [(1 - r^S) - (1 - r^B)] G^B \\ &= (1 - r^S) \Delta G + (r^B - r^S) G^B. \end{aligned}$$

Therefore:

$$\boxed{\Delta G_{h,t+1,k}^{\text{surv}} = \sum_a A_{a \rightarrow h,t} (1 - m_{a,t}) \left[(1 - r_{a,t,k}^S) \Delta G_{a,t,k} + G_{a,t,k}^B (r_{a,t,k}^B - r_{a,t,k}^S) \right].}$$

The first term inside the bracket carries the existing policy-induced delta forward, decayed by mortality and by realizations under the reform rate. The second term is *new* delta generated this year by differential realization behavior on baseline-stock: when $r^S < r^B$ the reform locks in additional gains, when $r^S > r^B$ the reform churns through baseline stock faster and shrinks future ΔG .

3.3 Death and Inheritance Flow with Regime Hooks

We now treat the disposition of the gain stock held by decedents at (a, t) , $m_{a,t} G_{a,t,k}^S$, and its baseline counterpart $m_{a,t} G_{a,t,k}^B$ (which always faces step-up — the baseline is current-law). Three policy regimes can apply on the reform side, distinguished by where decedent gains go:

Regime	Decedent gain stock under reform
step_up	Vanishes (gain is forgiven at death; no transfer, no realization).
carryover	Transfers to heirs with carryover basis; realized over time as heirs themselves realize.
deemed_realization	Forcibly realized on the decedent's final return at rate τ ; revenue accrues at death; nothing

We capture these three cases by a routing triple $(\delta_{\text{vanish}}, \delta_{\text{route}}, \delta_{\text{realize}})$ constrained to satisfy

$$\delta_{\text{vanish}} + \delta_{\text{route}} + \delta_{\text{realize}} = 1, \quad \delta_{\text{vanish}}, \delta_{\text{route}}, \delta_{\text{realize}} \in [0, 1],$$

with the canonical instantiations:

Regime	δ_{vanish}	δ_{route}	δ_{realize}
step_up	1	0	0
carryover	0	1	0
deemed_realization	0	0	1

The triple framing accommodates extension to mixed regimes (e.g., partial step-up with a carryover allowance above a threshold) as a future refinement without modifying the recurrence.

3.3.1 Stock contribution

Of the decedent gain stock, only the δ_{route} portion enters the next-period stock at heir cohorts. The reform-side contribution to $G_{h,t+1,k}^S$ from decedents is

$$G_{h,t+1,k}^{S,\text{death}} = \delta_{\text{route}} \sum_a \omega_{a \rightarrow h,t} m_{a,t} G_{a,t,k}^S.$$

The baseline-side contribution is zero, because baseline is step-up and $\delta_{\text{route}}^B = 0$. Therefore:

$$\Delta G_{h,t+1,k}^{\text{death}} = \delta_{\text{route}} \sum_a \omega_{a \rightarrow h,t} m_{a,t} (G_{a,t,k}^B + \Delta G_{a,t,k}).$$

This term has two parts. The first, $\delta_{\text{route}} \sum_a \omega \cdot m \cdot G^B$, is exogenous forcing: it is a level effect that arises purely from the regime change relative to current law's step-up, even with zero behavioral response. The second, $\delta_{\text{route}} \sum_a \omega \cdot m \cdot \Delta G$, is endogenous: it propagates whatever policy-induced delta was already in the decedent's cell to heir cells.

Within-cell derivation of the death stock. The expressions above use $m_{a,t}$ as a cell-level scalar, multiplying cell-total stock. That form silently assumes $\text{Cov}(m, G_{\text{unit}} \mid a, t) = 0$ — that within an age cell, individual mortality and individual gain stock are independent. This is empirically false: the wealth-mortality gradient makes the covariance large and negative (within any age cohort, the wealthier subpopulation both carries more accumulated gains and dies less). Using cell-mean m times cell-total G^B therefore overstates the true decedent stock contribution materially — by roughly $2.7\times$ at the gain-weighted aggregate in our data.

The correct calculation is per-record. The cell's decedent stock contribution is

$$D_{a,t} = \sum_i w_i m_i (G_{\text{unit},i} + dG_i),$$

where i indexes the tax units in the cell, w_i is the weight, m_i is per-record household mortality, $G_{\text{unit},i}$ is per-record unrealized gain stock, and dG_i is each record's share of the cell's policy-induced delta. Materializing per-record state in the recurrence is unnecessary: under the assumption that dG within a cell is allocated proportional to G_{unit} (each holder's share of the cell's gain stock — consistent with the distribution rule in §7.3), the sum collapses analytically:

$$\begin{aligned} D_{a,t} &= \sum_i w_i m_i G_{\text{unit},i} \cdot \left(1 + \frac{dG_a}{G_a^B}\right) \\ &= \left(\sum_i w_i m_i G_{\text{unit},i}\right) \cdot \frac{G_{a,t}^B + \Delta G_{a,t,k}}{G_{a,t}^B} \\ &= m_{a,t}^{\text{eff}} \cdot (G_{a,t,k}^B + \Delta G_{a,t,k}), \end{aligned}$$

where the effective cell mortality is the gain-weighted death rate

$$m_{a,t}^{\text{eff}} \equiv \frac{\sum_i w_i m_i G_{\text{unit},i}}{G_{a,t}^B}.$$

This m^{eff} is *not* an approximation — given the G -proportional allocation of dG , it is exactly the per-record sum. The same algebra applied to the survivor channel gives $(1 - m^{\text{eff}})$ multiplying the cell's surviving stock. Implementations should therefore substitute m^{eff} for m wherever m multiplies cell-aggregate stock in §3.2–3.3, with m^{eff} computed once per cell from the microdata as $\sum_i w_i m_i G_{\text{unit},i} / G_{a,t}^B$.

Choice of allocation rule. The G -proportional rule above is one of several reasonable choices. An alternative is to allocate dG proportional to positive realized gains:

$$dG_i^{(R)} = dG_a \cdot \frac{\max(kg_{it,i}, 0)}{R_a^B} \Rightarrow m_a^{\text{eff},R} = \frac{\sum_i w_i m_i \max(kg_{it,i}, 0)}{R_a^B},$$

which corresponds to the lock-in story (deferred realizations stay with the records that were going to realize the most). The choice between G and R rules is implementation-controlled rather than spec-fixed — the recurrence form $D = m^{\text{eff}} \cdot (G^B + \Delta G)$ is the same under either rule, with only the aggregate column changing. The simulator defaults to the G rule but exposes both via a configuration constant (`KG_DYN_DG_ALLOCATION` in `src/sim/kg_dynamics.R`); under R , the recurrence falls back to the G rate when $R_a^B = 0$ (e.g., young heir cohorts under carryover with no historical realization activity).

Practical impact. Step-up scenarios are unaffected: when $\delta_{\text{route}} = 0$ the death channel is shut off, and the $(1 - m)$ vs $(1 - m^{\text{eff}})$ misallocation in the survivor channel only shifts stock between "vanish at death" and "stay in the population," which are observationally equivalent under step-up. Carryover and deemed-realization scenarios change materially: per-record accounting removes the wealth-mortality bias that would otherwise inflate decedent stock by $\sim 2\text{--}3\times$, with proportional reductions in carryover inheritance flow and deemed-realization revenue.

3.3.2 Revenue contribution

The δ_{realize} portion of decedent gain stock is forcibly realized at death and contributes to revenue. Define the deemed-realization revenue accumulator:

$$R_{t,k}^{\text{death}} = \delta_{\text{realize}} \sum_a m_{a,t} (G_{a,t,k}^B + \Delta G_{a,t,k}) \bar{\tau}_{a,t},$$

where $\bar{\tau}_{a,t}$ is the cell-average capital gains MTR, possibly adjusted for any deemed-realization-specific rate or exemption. R^{death} is a parallel revenue stream, summed across asset classes and ages, that adds to the standard ΔR revenue developed in §8. Because baseline is step-up, baseline death-event capital-gains revenue is zero by construction; the level $R_{t,k}^{\text{death}}$ is therefore also the reform-vs-baseline delta.

The δ_{vanish} portion is what gives us no contribution to either next-period stock or revenue: gains forgiven at death exit the system.

3.4 Aging Transition Matrix and Topcoding

The aging matrix $A_{a \rightarrow h,t}$ allows the recurrence to be written uniformly across regular and topcoded ages. For a model with one-year age groups and a topcode at a_{max} :

$$A_{a \rightarrow h, t} = \begin{cases} 1 & \text{if } a < a_{\max} \text{ and } h = a + 1, \\ 1 & \text{if } a = a_{\max} \text{ and } h = a_{\max}, \\ 0 & \text{otherwise.} \end{cases}$$

This convention has two consequences. First, survivors at the topcode remain at the topcode; their delta accumulates within the topcode rather than aging out of the system. Second, the youngest age group $a_{\min}$ has no aged-survivor inflow and therefore $\Delta G_{a_{\min}, t+1, k}$ has no survivor contribution — only inheritance flow under carryover. Both boundary cases are handled mechanically by the matrix without requiring special-case logic in the recurrence.

3.5 Full Recurrence

Combining the survivor and death-event flows:

$$\boxed{\begin{aligned} \Delta G_{h, t+1, k} &= \sum_a A_{a \rightarrow h, t} (1 - m_{a, t}) \left[(1 - r_{a, t, k}^S) \Delta G_{a, t, k} + G_{a, t, k}^B (r_{a, t, k}^B - r_{a, t, k}^S) \right] \\ &\quad + \delta_{\text{route}} \sum_a \omega_{a \rightarrow h, t} m_{a, t} (G_{a, t, k}^B + \Delta G_{a, t, k}). \end{aligned}}$$

This is the central law of motion for the policy-induced delta in unrealized capital gains. It is linear in ΔG , applies cell-by-cell with no within-year recursion (given $\Delta G_{a, t, k}$ for all cells, the year- $(t+1)$ values for all cells follow from a single pass), and reduces to zero under reform-equals-baseline (since $r^S = r^B$ and $\delta_{\text{route}} = 0$ collapses both terms).

3.6 Aggregate (Vector) Form

Stacking $\Delta G_{a, t, k}$ over a into a column vector $\mathbf{\Delta G}_{t, k} \in \mathbb{R}^{|a|}$, define:

- The diagonal "own-cell decay" matrix $D_t^{(k)} \in \mathbb{R}^{|a| \times |a|}$ with entries $D_t^{(k)}[a, a] = (1 - m_{a, t})(1 - r_{a, t, k}^S)$.
- The aging matrix $A_t \in \mathbb{R}^{|a| \times |a|}$ acting on the destination index.
- The heir-routing matrix $W_t \in \mathbb{R}^{|a| \times |a|}$ with entries $W_t[a, h] = \omega_{a \rightarrow h, t} m_{a, t}$.
- The exogenous forcing vector $\mathbf{f}_t^{(k)} \in \mathbb{R}^{|a|}$ with entries $f_{t, h}^{(k)} = \sum_a A_{a \rightarrow h, t} (1 - m_{a, t}) G_{a, t, k}^B (r_{a, t, k}^B - r_{a, t, k}^S) + \delta_{\text{route}} \sum_a W_t[a, h] G_{a, t, k}^B$.

Then the recurrence reads compactly:

$$\mathbf{\Delta G}_{t+1, k} = (A_t^\top D_t^{(k)} + \delta_{\text{route}} W_t^\top) \mathbf{\Delta G}_{t, k} + \mathbf{f}_t^{(k)}.$$

This is a linear time-varying system. The transition operator $A_t^\top D_t^{(k)} + \delta_{\text{route}} W_t^\top$ is dissipative under step-up ($\delta_{\text{route}} = 0$, since the diagonal decay multipliers are strictly less than one) and quasi-conservative under carryover (the heir-routing term redistributes magnitude between cells). Sensitivity of $\mathbf{\Delta G}$ to parameter perturbations propagates linearly.

3.7 Conservation Properties

Three conservation statements follow directly from the recurrence:

1. **Carryover conserves routed magnitude at the transfer event.** Under $\delta_{\text{route}} = 1$, the total decedent contribution $\sum_a m_{a,t} (G_{a,t,k}^B + \Delta G_{a,t,k})$ is exactly the total inheritance inflow $\sum_h \sum_a \omega_{a \rightarrow h,t} m_{a,t} (G^B + \Delta G)$, by row-stochasticity of ω . The death-event transfer itself is conservative; aggregate ΔG over the full simulation horizon still evolves through survivor decay $(1 - m)(1 - r^S)$ and differential realizations $(r^B - r^S) G^B$, so this is not a statement about long-run conservation of ΔG .
 1. **Deemed converts disposed Δ into revenue.** Under $\delta_{\text{realize}} = 1$, the decedent stock contributes nothing to next-period ΔG but contributes R^{death} to revenue. The two channels are complementary: stock disposed equals revenue raised, modulo the rate.
 1. **Step-up dissipates Δ .** Under $\delta_{\text{vanish}} = 1$, decedent stock leaves the system without replacement. The bathtub slowly leaks through mortality even with no realization activity.
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4. Realization Rate Function

The reform-side realization rate is derived from a **representative-agent choice problem**: in each period the holder decides what fraction of their embedded gain to realize, trading off non-tax benefits of liquidity against the *effective tax price* of realizing now versus deferring. The regime (step-up, carryover, deemed) enters through a single object — the effective tax price P — which propagates simultaneously into the *level* of realizations and into their sensitivity to the tax rate. This eliminates the need for separate "level" and "elasticity" parameters and ties their values to the same underlying mechanism.

4.1 The agent's choice problem

Normalize to \$1 of unrealized gain. A representative holder in cell (a, t, k) picks a realization fraction $r \in [0, 1]$ to maximize:

$$\max_{r \in [0,1]} b_{a,k}(r) - P_{a,t,k} \cdot r,$$

where $b_{a,k}(r)$ is a concave non-tax benefit function (capturing liquidity needs, rebalancing motives, and other reasons to realize that are unrelated to taxes), and $P_{a,t,k}$ is the effective tax price of realizing now rather than holding (developed in §4.2).

For tractability we choose the marginal-benefit form $b'(r) = \mu_{a,k} - (1/\eta_k) \log r$, which gives the FOC $\log r^* = \eta_k(\mu_{a,k} - P)$. The cell-specific intercept $\mu_{a,k}$ absorbs all non-tax heterogeneity in realization motives and is implicitly calibrated by the baseline anchoring (§5). The free parameter is η_k , the curvature of the marginal-benefit curve, which controls how sensitively the holder reshuffles realizations in response to changes in the tax price.

Differencing across baseline and reform and exponentiating yields the **baseline-anchored realization rate**:

$$r_{a,t,k}^S = r_{a,t,k}^B \cdot \exp(-\eta_k \cdot (P_{a,t,k}^S - P_{a,t,k}^B)).$$

When the reform coincides with baseline ($P^S = P^B$), the exponential collapses to one and $r^S = r^B$ exactly, guaranteeing $\Delta G \equiv 0$ when no policy change is in effect.

4.2 Effective tax price

If the holder sells today, they pay τ_t per dollar of gain. If they hold, they may sell voluntarily later, die before realizing, have heirs inherit a tax burden under carryover, or face a deemed realization at death. The effective tax price is the wedge between paying-now and the present value of the expected future tax payment if they hold:

$$P_{a,t,k}(c) = \tau_t - \sum_{j=1}^{\infty} \beta^j s_{a,k,j} \tau_{t+j} - \sum_{j=1}^{\infty} \beta^j d_{a,j} c \tau_{t+j},$$

where j indexes future years and:

- $s_{a,k,j}$ — probability that the asset is voluntarily realized in year $t + j$, conditional on the holder still holding at t . Built from the competing-risks recursion in §4.3.
- $d_{a,j}$ — probability that the holder dies in year $t + j$ before voluntarily realizing, again from the §4.3 recursion.
- β — annual discount factor (≈ 0.96).
- τ_{t+j} — the path of expected future tax rates. Under naive expectations, $\tau_{t+j} = \tau_t$ for all j ; under announced rate changes, τ_{t+j} tracks the announced path.
- $c \in [0, 1]$ — the regime hook from §2.4. The fraction of the embedded tax burden that survives the holder's death and is internalized in their lifetime decision.

The sums are written formally to infinity; the integrand decays geometrically at rate $\beta \cdot (1 - \lambda_r - m) \approx 0.82$ per year, so terms are below float precision after a few hundred years. The implementation truncates at 200 years, which is well into the irrelevant tail and exposes no tunable parameter — β alone governs how quickly future events are discounted away.

The first sum is the discounted expected tax from voluntary future realization. The second sum is the discounted expected tax from the death state, scaled by the regime's burden-share c . Under step-up ($c = 0$) the death state contributes nothing — gains are forgiven. Under deemed realization ($c = 1$) the full tax is paid at death. Under carryover, c captures the holder's bequest motive: $c = 1$ means the heir's eventual tax is fully internalized; $c = 0$ collapses to step-up.

Under constant rate expectations ($\tau_{t+j} = \tau_t$), P factors cleanly:

$$P_{a,t,k}(c) = \tau_t \cdot (1 - M_{a,k}(c)), \quad M_{a,k}(c) = \sum_{j=1}^{\infty} \beta^j s_{a,k,j} + c \sum_{j=1}^{\infty} \beta^j d_{a,j}.$$

The dimensionless quantity $1 - M_{a,k}(c)$ is the **bracket**. It scales both the level of P and the rate sensitivity (§4.4).

4.3 Hazard sequences from competing risks

The probabilities s_j and d_j come from a discrete-time competing-risks process. Each year, conditional on still holding, the holder either voluntarily realizes (with hazard $\lambda_{r,k}$, the per-period realization hazard for asset class k), dies (with cell-mortality $m_{a,t}$), or holds another year. Stacking forward:

$$S_{a,j} = \prod_{i=0}^{j-1} (1 - \lambda_{r,k} - m_{a+i,t+i}), \quad S_{a,0} = 1,$$

is the probability that the asset is *still being held* at the start of year $t + j$. Then:

$$s_{a,k,j} = S_{a,j} \cdot \lambda_{r,k}, \quad d_{a,j} = S_{a,j} \cdot m_{a+j,t+j}.$$

The indexing pairs s_j and d_j with the bracket-sum factors β^j and τ_{t+j} in §4.2: realizing or dying *in year $t + j$* requires the holder to have survived from year t through the start of year $t + j$ (probability $S_{a,j}$, j years of competing-risks decay), and then to realize / die during year $t + j$ at age $a + j$.

For ages above the topcode $a_{\max}$, m is held fixed at $m_{a_{\max}}$. The mortality path $m_{a+i,t+i}$ is read from the simulator's per-year cell aggregates or from a fixed life table.

The realization hazard $\lambda_{r,k}$ is the only *new* parameter introduced by the microfoundation and is calibrated from observed baseline realizations (§6.4).

4.4 Properties

Level effect: removing step-up raises the level of realizations. Going from $c = 0$ (step-up) to $c > 0$ (carryover, deemed) makes the second sum in §4.2 nonzero, *raising* expected future tax burden, *lowering* P , and therefore *raising* r^S . The size of the increase scales with mortality — holders with high m see the biggest reduction in P because the death state contributes most to their expected future tax. Young holders see nearly no change. This reproduces the "elderly lock-in unwinds when step-up is removed" property without a separate ψ parameter.

Sensitivity effect: removing step-up reduces tax-rate elasticity. Differentiating P with respect to τ :

$$\frac{\partial P_{a,t,k}}{\partial \tau_t} = 1 - \sum_j \beta^j s_j \frac{\tau_{t+j}}{\tau_t} - c \sum_j \beta^j d_j \frac{\tau_{t+j}}{\tau_t} = 1 - M_{a,k}(c) \quad (\text{under constant rate expectations}).$$

The semi-elasticity of r^S with respect to τ is therefore:

$$\boxed{\frac{\partial \log r^S}{\partial \tau_t} = -\eta_k \cdot (1 - M_{a,k}(c)).}$$

The same bracket $1 - M_{a,k}(c)$ that sets the level of P also sets the slope. Under deemed ($c = 1$) the bracket is small (most of a rate hike is "passed through" into expected future tax burden, neutralizing the wedge change), so the rate elasticity is small. Under step-up ($c = 0$) the bracket is larger (the rate-hike-induced increase in expected future tax is weighted only by the survival probability $1 - M_{\text{die}}$, since the death state pays nothing), so the rate elasticity is larger. **The level bump and the elasticity reduction are two faces of a single mechanism: the bracket.**

Forward-looking transitions emerge naturally. If a rate change is announced for year t^* , the sums in §4.2 use the announced path τ_{t+j} , which causes P — and therefore r^S — to respond *before* the change takes effect. Anticipation behavior is captured without modifying the recurrence. The model remains myopic about *unannounced* future changes, but accommodates expected paths.

Transitory release on rate cuts emerges from the dynamics, not the rate function. Identical mechanism to the previous reduced-form: at the moment of a rate cut, accumulated $\Delta G > 0$ from prior lock-in is hit with a rising r^S , producing a surge of realizations beyond the steady-state response. The bracket framework does not change this; it is a property of how ΔG is multiplied through the §3.5 recurrence.

4.5 Connection to reduced-form realization equations

Setting $P = P^B + \tilde{\varepsilon} \cdot (\tau - \tau^B)$ and ignoring the second-sum dependence on τ recovers the log-linear form $\log r^S = \log r^B - \tilde{\varepsilon} \cdot \eta \cdot (\tau^S - \tau^B)$ that Burman-Randolph (1994), Dowd-McClelland-Muthitacharoen (2015), and similar reduced-form realization equations adopt. The microfoundation makes explicit that the literature's "semi-elasticity" $\tilde{\varepsilon}$ implicitly bundles together the agent's behavioral curvature (η), the horizon over which they expect to hold (β raised to the expected-time-to-realization), and the regime (c). Decomposing into $(\eta, \beta, \lambda_r, c)$ exposes the implicit assumptions and lets the analyst vary them independently.

5. Empirical Anchoring

5.1 The ratio formulation

The realization function r specified in §4 is a structural object. In application, the model must be anchored so that it reproduces observed baseline realizations exactly when there is no reform; calibration imperfections in r should affect *differences* between scenarios but not baseline levels. We achieve this by applying r via a ratio to the observed baseline:

$$\text{realizations}^S(a, t, k) = \text{realizations}_{\text{obs}}^B(a, t, k) \cdot \frac{r_{a,t,k}^S G_{a,t,k}^S}{r_{a,t,k}^B G_{a,t,k}^B},$$

where $\text{realizations}_{\text{obs}}^B$ is the simulator's baseline realized-gain output for the cell.

When the scenario is itself baseline, $r^S = r^B$ and $G^S = G^B$, the ratio collapses to one, and $\text{realizations}^S = \text{realizations}_{\text{obs}}^B$ identically. With the baseline-anchored form of r^S in §4.1, this is automatic — no additional miscalibration of r is possible to break it.

Sparse-cell guard. The ratio is undefined when $r_{a,t,k}^B G_{a,t,k}^B = 0$. In such cells (typically young heir cohorts that have just received carryover-transferred gains but have no historical realization

activity), the ratio formula is replaced by a fallback rule: apply $r^S \cdot G^S$ directly, or pool r^B across age within the asset class, depending on which fallback the simulator’s sparse-cell convention dictates. This is the same hierarchy used by the within-cell allocator in §7.3.

5.2 Equivalence under perfect calibration

Under perfect calibration of r , the ratio formulation is mathematically equivalent to applying r^S directly to G^S . The two differ only when r^B as computed by the structural formula does not equal the observed baseline rate. The ratio formulation is robust to this discrepancy; the direct-level form is not.

5.3 Reform-induced delta in realizations

The flow object that feeds revenue is

$$\Delta R_{a,t,k} = \text{realizations}^S(a, t, k) - \text{realizations}_{\text{obs}}^B(a, t, k).$$

Algebraically:

$$\Delta R_{a,t,k} = \text{realizations}_{\text{obs}}^B(a, t, k) \left[\frac{r^S (G^B + \Delta G)}{r^B G^B} - 1 \right].$$

This is the cell-aggregate ΔR . Distribution to individual tax units within the cell follows the convention in §7.3.

6. Calibration

The microfoundation in §4 introduces five parameter groups, each with a distinct identification strategy.

6.1 Baseline rates $r_{a,t,k}^B$ — observed

The baseline realization rate by age, year, and asset class is taken directly from the simulator’s baseline pass. It is not estimated; it is not a free parameter. The simulator’s baseline aggregates realized gains by age and asset class and divides by the cell’s baseline gain stock to produce a rate profile.

Baseline reproduction is built into the realization function via the exponential anchoring in §4.1: when reform equals baseline, $P^S = P^B$ and $r^S = r^B$ exactly, regardless of how η_k or any other parameter is calibrated.

6.2 Behavioral curvature η_k — calibrated

The curvature of the marginal-benefit-of-realizing curve, η_k , controls how strongly r^S responds to changes in the effective tax price. It is calibrated so that the model’s implied aggregate realization elasticity matches a literature target under current law (step-up).

Procedure. Define the implied semi-elasticity at the baseline rate:

$$\left. \frac{\partial \log \sum_{a,t} \text{realizations}^S(a, t, k)}{\partial \tau} \right|_{\tau=\tau_0, c=0} = -\eta_k \cdot (1 - M_{a,k}(0))_{\text{aggregated over } a, t}.$$

Choose a target value from the realization elasticity literature (typically $\eta \approx -0.5$ to -1.0 for equities; smaller magnitudes for less liquid assets) and solve for the unique η_k that produces it under $c = 0$. The bracket $1 - M_{a,k}(0)$ depends on the cell's mortality path and the realization hazard $\lambda_{r,k}$ (§6.4), both of which are pinned down before η_k is calibrated.

Existence and uniqueness are straightforward: the implied elasticity is linear in η_k holding the bracket fixed.

6.3 Discount factor β — empirical

The discount factor β is read from financial-economics estimates of household discount rates in the lock-in literature, typically $\beta \in [0.95, 0.97]$ corresponding to a 3–5% annual discount rate. Auerbach (1991) and Constantinides (1983) use $\beta \approx 0.96$. The doc adopts $\beta = 0.96$ as the default and treats it as common across asset classes.

For greater rigor, β could be made asset-specific (housing carries a different effective opportunity cost than equities) or risk-adjusted; the minimal model does not pursue this.

6.4 Realization hazard $\lambda_{r,k}$ — calibrated from data

The voluntary realization hazard for asset class k is calibrated so that the §4.3 competing-risks recursion reproduces the observed aggregate realization rate at baseline. Concretely:

$$\lambda_{r,k}^{\text{calibrated}} = \frac{\sum_{a,t} R_{a,t,k}^B}{\sum_{a,t} G_{a,t,k}^B},$$

i.e., the asset-class-aggregate ratio of realized gains to unrealized stock at baseline. This is a single number per asset class (not per cell) and serves as the "average" hazard the holder uses to forecast their own future realization timing. Typical values: $\lambda_r \approx 0.05$ for liquid equities, 0.02 for housing.

This is the simplest possible calibration. Refinements could vary $\lambda_{r,k}$ by age or wealth, but at the cost of losing the parsimony.

6.5 Regime burden share c — judgmental, regime-specific

The regime hook $c \in [0, 1]$ is set by policy:

Regime	c
step_up	0
carryover	$\theta \in (0, 1]$ — bequest-motive parameter
deemed_realization	1

Step-up and deemed realization pin c at the endpoints; carryover requires the bequest-motive parameter θ . We treat θ as a tunable assumption — there is no domestic empirical anchor because carryover has never been enacted in the modern U.S. — with default $\theta = 0.5$ (the holder internalizes

half of the heir’s eventual tax burden). Sensitivity to θ should be reported at ± 0.25 around the default.

Note that θ is the *only* tunable parameter for which the model has no empirical anchor. The previous reduced-form spec had two (ε calibrated; ψ tunable). The microfoundation pins down the analog of ψ via the bracket calculation, leaving only θ as a knob — a substantial improvement in identification.

6.6 Inheritance allocation matrix ω — exogenous

Unchanged from the reduced-form spec: $\omega_{a \rightarrow h, t}$ is supplied by the estate module and treated here as exogenous, with the only constraint being row-stochasticity in h . The estate module determines the heir age distribution conditional on decedent age. ω is regime-indifferent in its construction; its effect on the bathtub is gated by the regime hook δ_{route} in §3.3.

6.7 Summary of identification

Parameter	Cardinality	Identification
$r_B(a, k)$	$\ a\ \cdot \ \mathcal{K}\ $	Observed from simulator baseline
$m(a, t)$	$\ a\ $	Observed from cell aggregates / life table
η_k	$\ \mathcal{K}\ = 5$	Calibrated to step-up elasticity target
$\lambda_{r, k}$	$\ \mathcal{K}\ = 5$	Calibrated from observed R^B/G^B
β	1	Empirical default ≈ 0.96
θ (bequest motive)	1 (or per- k)	Tunable, default 0.5, sensitivity panel
ω	exogenous	From estate module

Total tunable-or-calibrated parameters: $2|\mathcal{K}| + 2 = 12$ for five asset classes (five η_k + five $\lambda_{r, k}$ + β + θ). Of these, ten are *calibrated* against observed quantities (η_k from elasticity targets, $\lambda_{r, k}$ from baseline data); only β and θ are judgmental.

In comparison, the reduced-form spec had ten free parameters (five ε_k + five ψ_k), of which five were calibrated and five were tunable. The microfoundation has slightly more parameters but a much higher fraction of them are anchored to observable quantities, and the level-vs-elasticity tradeoff that motivated separate (ε, ψ) in the reduced form is now an internally consistent consequence of a single mechanism.

7. Conventions and Boundaries

7.1 Joint-filer cohort assignment

Joint-filer households are assigned a cohort using the older spouse’s age:

$$a_{\text{joint}} = \max(a_1, a_2).$$

This is the natural anchor: the older spouse is more likely to be near end-of-life, which is when the step-up benefit binds most strongly, and is typically the asset-side anchor in joint holdings. The household survival probability is

$$m_{\text{joint},t} = m_{1,t} \cdot m_{2,t},$$

i.e., the household ceases to exist as a joint-filer cell only when both spouses die in the same year. Widow continuation is handled by the cross-section's own re-imputation in subsequent years; the surviving spouse appears in their own age cell with their own assets.

7.2 Boundary cells

- **Youngest cohort.** Tax units at $a_{\min}$ have no aged-survivor inflow by construction. Their ΔG accumulates only through inheritance flow under carryover. Realistically, $a_{\min} = 18$ or 20 in U.S. microsim contexts, by which age inherited gains become possible.
- **Topcoded cohort.** At $a_{\max}$, the aging matrix $A_{a_{\max} \rightarrow a_{\max},t} = 1$ retains survivors within the topcode. Mortality at the topcode is high and effectively dominates the cell's survivor decay, but the recurrence remains well-defined and stable as long as $0 \leq m \leq 1$.

7.3 Within-cell distribution to tax units

The recurrence and the realization function operate on cell aggregates. The distribution of cell-aggregate $\Delta R_{a,t,k}$ to individual tax units within the cell uses the following rule:

1. **Pro-rata to baseline realizations.** If the cell has nonzero baseline realized gains, each tax unit's share of ΔR equals its share of the cell's baseline realizations.
1. **Pro-rata to baseline gain stock.** Fall back to the share of G^B when the cell has zero baseline realizations (e.g., a young cell that has just received carryover-transferred gains but has historically not sold).
1. **Skip.** If both baseline realizations and G^B are zero in the cell, no realizations are distributed; the cell contributes nothing to revenue that year.

The fallback hierarchy is needed in carryover scenarios because young heir cohorts can accumulate ΔG before they have any baseline realization activity to anchor against.

8. Revenue and Downstream Use

8.1 Ongoing realization revenue

For each (a, t, k) , the cell-level revenue contributions under reform and baseline are

$$T_{a,t,k}^S = \tau_{a,t}^S \cdot R_{a,t,k}^S, \quad T_{a,t,k}^B = \tau_{a,t}^B \cdot R_{a,t,k}^B,$$

where $R^S \equiv \text{realizations}^S$ from §5 and R^B is observed baseline realizations. The reform-vs-baseline delta is

$$\boxed{\Delta T_{a,t,k} = \tau_{a,t}^S R_{a,t,k}^S - \tau_{a,t}^B R_{a,t,k}^B.}$$

This general form is necessary because policy reforms typically change both the rate τ and the realization base R . When the rate is unchanged ($\tau^S = \tau^B$), the expression simplifies to $\Delta T = \tau \cdot \Delta R$; when only the rate changes ($R^S = R^B$), it simplifies to $\Delta T = (\tau^S - \tau^B) R$. Mixed reforms exhibit both channels and the cross-term cannot be separated cleanly into rate and base effects.

8.2 Death-event revenue under deemed realization

Under deemed realization, the additional revenue stream is

$$R_{t,k}^{\text{death}} = \delta_{\text{realize}} \sum_a m_{a,t} (G_{a,t,k}^B + \Delta G_{a,t,k}) \bar{\tau}_{a,t},$$

as in §3.3.2. This is computed in parallel to the standard ΔT stream and aggregated to the year-asset level.

8.3 Total revenue impact

The total reform-vs-baseline revenue impact for year t is

$$\Delta T_t = \sum_{a,k} \Delta T_{a,t,k} + \sum_k R_{t,k}^{\text{death}},$$

with the first term active under all regimes and the second active only under deemed realization. This is the object that integrates into the simulator's existing revenue pipeline.

9. Acknowledged Gaps

The model leaves several phenomena outside its scope. Each is documented here to make the modeling choices explicit.

- **Charitable dispositions of appreciated assets.** Donating appreciated stock allows the donor to deduct fair market value without realizing the embedded gain. This bleeds the unrealized stock without going through the realization rate function. The model treats this as a calibration adjustment to G^B (downward bias) rather than an explicit channel.
- **Forward-looking expectations.** The §4 framework accommodates *announced* future rate paths: τ_{t+j} in the effective-tax-price formula can be set to a known forward path, in which case anticipation effects emerge naturally (holders pre-realize before announced hikes, defer before announced cuts). What remains out of scope is rational expectations of *unannounced* future changes — i.e., the model does not endogenously forecast future tax-policy uncertainty. Under naive expectations $\tau_{t+j} = \tau_t$, the model collapses back to a myopic-but-multi-period forward-looking formulation. Full rational expectations would require a Bellman solve over the value of holding, which the doc deliberately avoids.
- **Mark-to-market regimes.** Annual taxation of accruals collapses ΔG to near-zero by construction (all appreciation is realized contemporaneously). The bathtub plumbing remains valid but the channel through which revenue accrues differs fundamentally. Out of scope here.

- **Accruals as a policy-dependent flow.** This document treats accruals on existing assets as policy-invariant. Wealth feedback effects (deemed realization shrinks estates, smaller estates accrue smaller future gains) are second-order and not modeled. Revenue from the deemed channel is captured directly; the wealth-feedback channel is acknowledged and deferred.
 - **Sparse-cell calibration.** Some (a, k) cells have thin baseline observations (e.g., young cohorts holding pass-through stakes). The ratio anchoring still applies but r_B may be noisy. Default fallback is to pool r_B across age within an asset class when sample size in a cell falls below a threshold.
-

10. Implementation Notes

The implementation maps the formalism to the simulator’s existing architecture. This section is light on detail by intent; it points to the correct hook points and patterns rather than specifying code.

10.1 Hook points

- **Per-year cross-sectional state.** The state vector $\Delta \mathbf{G}_{t,k}$ for each scenario $\times$ asset class is persisted to disk between years, mirroring the disk-based across-year state pattern already used elsewhere in the simulator for behavioral feedback that requires prior-year context.
- **Conventional pass only.** The bathtub state evolves only under the conventional simulation pass. The static pass holds inputs at baseline and produces $\Delta G \equiv 0$ by construction.
- **Year sequencing.** Year-level parallelization is incompatible with the recurrence: year $t + 1$ requires year t ’s state. Simulations using this module must run years sequentially within a scenario; parallelization across scenarios remains available.

10.2 Configuration

- **Tax-law parameters.** The regime burden share c and the routing triple $(\delta_{\text{vanish}}, \delta_{\text{route}}, \delta_{\text{realize}})$ are policy choices and live in the tax-law YAML hierarchy alongside the rate schedule. A new file (e.g., `death.yaml`) is the natural home.
- **Behavioral parameters.** η_k , β , $\lambda_{r,k}$, and θ (bequest motive) are calibration constants, not policy choices. They live in a separate behavioral-parameters file outside the tax-law tree, alongside other model fixtures such as mortality tables.
- **Pre-computed brackets.** The bracket $1 - M_{a,k}(c)$ depends only on the life-table mortality path, $\lambda_{r,k}$, β , and c — not on the simulation year or on ΔG . It is computed once at scenario startup as a per-cell, per-regime constant and reused across all years.
- **Inheritance allocation matrix.** ω is supplied by the estate module’s per-year output, extended to carry decedent-age and heir-age cohort markers and per-asset transfer amounts.

10.3 Pseudocode

```

# at scenario startup
for each asset class k, each age a:
    bracket_B[a,k] = compute_bracket(a, c=0, m_path, λ_r[k], β)
    bracket_S[a,k] = compute_bracket(a, c=c_reform, m_path, λ_r[k], β)
    P_B[a,k]       = τ_B · (1 - bracket_B[a,k])
    P_S[a,k]       = τ_S · (1 - bracket_S[a,k])

for each year t in simulation_horizon:
    Δ_prev = load_cohort_state(scenario_id, t) # zero in first year

    for each asset class k:
        r_S[a,t,k] = r_B[a,t,k] · exp( -η[k] · (P_S[a,k] - P_B[a,k]) )

        for each destination age h:
            survivor = Σ_a A[a→h] · (1 - m[a,t])
                      · [ (1 - r_S[a,t,k]) · Δ_prev[a,k]
                          + G_B[a,t,k] · (r_B[a,t,k] - r_S[a,t,k]) ]
            inherit  = δ_route · Σ_a ω[a→h,t] · m[a,t]
                      · (G_B[a,t,k] + Δ_prev[a,k])
            Δ_next[h,k] = survivor + inherit

        if δ_realize > 0:
            R_death[t,k] = δ_realize · Σ_a m[a,t]
                          · (G_B[a,t,k] + Δ_prev[a,k]) · τ[a,t]

    realizations_S = baseline_obs · (r_S · G_S) / (r_B · G_B)
    distribute_within_cell(realizations_S - baseline_obs) # pro-rata
    ΔT[t] = Σ_a,k τ[a,t] · ΔR[a,t,k] + Σ_k R_death[t,k]

    persist_cohort_state(scenario_id, t+1, Δ_next)

```

10.4 Migration from existing aggregate-elasticity modules

The existing capital-gains behavioral feedback modules apply a single aggregate elasticity to realized gains uniformly. This framework subsumes those modules: the cohort $\times$ asset specification, the regime-conditional death routing, and the empirical anchoring to baseline together produce a strictly richer and more disciplined model of realization behavior. Activation of this framework retires the legacy aggregate modules. Existing runscripts that toggled the legacy modules continue to function under default settings ($c = 0$, i.e., step-up baseline) but should be reviewed for consistency with the calibrated η_k , β , $\lambda_{r,k}$ parameter set.